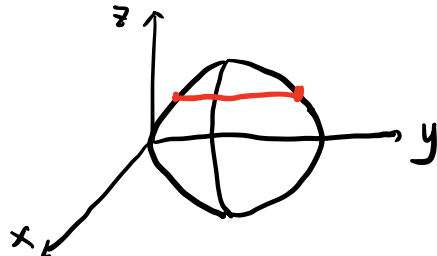


Lecture 14 Triple Integrals Cont'd

Recap: $D, f(x, y, z)$

$$\iiint_D f(x, y, z) dV \begin{cases} \text{by pillars} \\ \text{by slices} \end{cases}$$

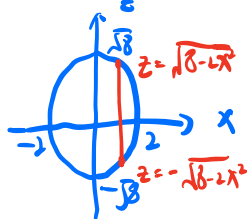
Example. Let D be the solid bounded by the paraboloid $y = x^2 + z^2$ and $y = 16 - 3x^2 - z^2$. Find the volume of the solid.



$$x^2 + z^2 = 16 - 3x^2 - z^2$$

$$4x^2 + 2z^2 = 16$$

$$2x^2 + z^2 = 8$$



$$x = 2 \sin \theta$$

$$dx = 2 \cos \theta d\theta$$

$$\cos^2 \theta = \frac{1 + \cos 2\theta}{2}$$

$$\cos^4 \theta = \frac{1}{4} + \frac{1}{2} \cos 2\theta + \frac{1}{4} \frac{1 + \cos 4\theta}{2}$$

$$= \frac{3}{8} + \frac{1}{2} \cos 2\theta + \frac{1}{8} \cos 4\theta$$

Sol. $\iiint_D dV$

$$= \int_{x=-2}^{x=2} \int_{z=-\sqrt{8-2x^2}}^{z=\sqrt{8-2x^2}} \int_{y=x^2+z^2}^{y=16-3x^2-z^2} dy dz dx$$

$$= \int_{-2}^2 \int_{-\sqrt{8-2x^2}}^{\sqrt{8-2x^2}} (16 - 4x^2 - 2z^2) dz dx$$

$$= \int_{-2}^2 \left[16z - 4x^2 z - \frac{2}{3} z^3 \right] \Big|_{-\sqrt{8-2x^2}}^{\sqrt{8-2x^2}} dx$$

$$= \int_{-2}^2 \left(32\sqrt{8-2x^2} - 8x^2\sqrt{8-2x^2} - \frac{4}{3}(8-2x^2)\sqrt{8-2x^2} \right) dx$$

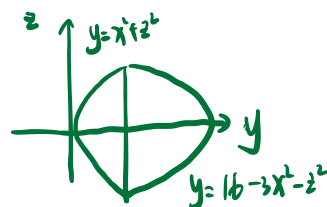
$$= \int_{-2}^2 \frac{8}{3}(8-2x^2)\sqrt{8-2x^2} dx$$

$$= \frac{16}{3} \int_0^2 (8-2x^2)\sqrt{8-2x^2} dx$$

$$= \frac{16}{3} \int_0^{\frac{\pi}{2}} 8 \cos^2 \theta \sqrt{8} \cos \theta \cdot 2 \cos \theta d\theta$$

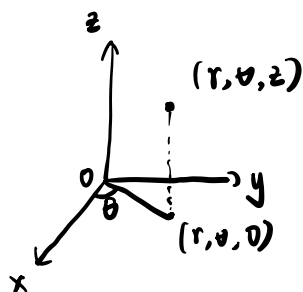
$$= \frac{16}{3} 32\sqrt{2} \int_0^{\frac{\pi}{2}} \cos^4 \theta d\theta$$

$$= \frac{16}{3} 32\sqrt{2} \frac{3}{8} \frac{\pi}{2} = 32\pi\sqrt{2}$$



Cylindrical coordinates

$$\begin{cases} x = r \cos \theta \\ y = r \sin \theta \\ z = z \end{cases}$$



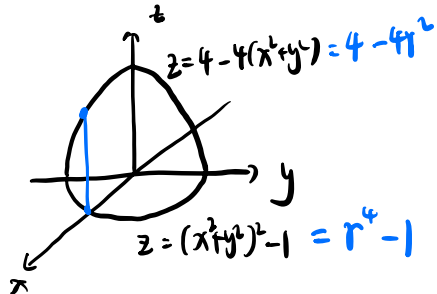
$r = \text{constant}$ represents an infinite cylinder in $\mathbb{R}^3$.

Theorem. Under cylindrical coordinates (r, θ, z) , we have

$$dV = r dz dr d\theta$$

Example. Find the volume of the solid bounded by two surfaces $z = 4 - 4(x^2 + y^2)$

and $z = (x^2 + y^2)^2 - 1$.



Sol. $4 - 4r^2 = r^4 - 1$

$$r^4 + 4r^2 - 5 = 0$$

$$(r^2 + 5)(r^2 - 1) = 0$$

$r = 1 \Rightarrow z = 0$ the intersection is the unit circle on the xy plane

$$\iiint_D dV = \int_0^{2\pi} \int_0^1 \int_{r^4-1}^{4-4r^2} r dz dr d\theta.$$

$$= \int_0^{2\pi} \int_0^1 r(4 - 4r^2 - r^4 + 1) dr d\theta$$

$$= \int_0^{2\pi} d\theta \cdot \int_0^1 (5r - 4r^3 - r^5) dr$$

$$= 2\pi \left[\frac{5r^2}{2} - r^4 - \frac{r^6}{6} \right] \Big|_0^1$$

$$= \frac{8\pi}{3}.$$

Change of variables

Suppose each coordinate of (x, y, z) is a function of (u, v, w)

$$\text{Jacobian matrix: } \frac{\partial(x, y, z)}{\partial(u, v, w)} = \begin{pmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} & \frac{\partial x}{\partial w} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} & \frac{\partial y}{\partial w} \\ \frac{\partial z}{\partial u} & \frac{\partial z}{\partial v} & \frac{\partial z}{\partial w} \end{pmatrix}$$

$$dV = dx dy dz = \left| \det \frac{\partial(x, y, z)}{\partial(u, v, w)} \right| du dv dw.$$

$$x = r \cos \theta, \quad y = r \sin \theta, \quad z = z$$

$$\frac{\partial(x, y, z)}{\partial(r, \theta, z)} = \begin{pmatrix} \cos \theta & -r \sin \theta & 0 \\ \sin \theta & r \cos \theta & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$\det \frac{\partial(x, y, z)}{\partial(r, \theta, z)} = \det \begin{pmatrix} \cos \theta & -r \sin \theta \\ \sin \theta & r \cos \theta \end{pmatrix} = r \cos^2 \theta + r \sin^2 \theta = r$$

$$dV = r dr d\theta dz.$$

Example. Let D be a solid cylinder of radius a with z -axis as the central axis, is bounded by planes $z = z_0$ and $z = z_0 + h$. Suppose the solid has uniform density δ . Evaluate the following trip integral:

$$I_z = \iiint_D \delta(x^2 + y^2) dV. \quad \text{moment of inertia.}$$

Sol. $r \leq a, \quad 0 \leq \theta \leq 2\pi, \quad z_0 \leq z \leq z_0 + h$

$$I_z = \int_{z_0}^{z_0+h} \int_0^{2\pi} \int_0^a \delta r^2 r dr d\theta dz.$$

$$= \left(\int_{z_0}^{z_0+h} dz \right) \cdot \left(\int_0^{2\pi} d\theta \right) \cdot \left(\int_0^a \delta r^3 dr \right)$$

$$= h \cdot 2\pi \delta \cdot \left[\frac{r^4}{4} \right]_0^a$$

$$= h \cdot 2\pi \cdot \delta \cdot \frac{a^4}{4}$$

$$= \frac{sh\pi a^4}{2}$$

$$\delta = \frac{m}{V} = \frac{m}{\pi a^2 h}$$

$$I_z = \frac{m}{\pi a^2 h} \cdot \frac{h\pi a^4}{2} = \frac{1}{2} m a^2.$$